INTRODUCTION TO E-COMMERCE

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Slides based on chapters 1 & 3 of Laurdon & Traver e- commerce book

Learning Objectives

- **Define e-commerce and describe how it differs from e-business**
- **Identify the unique features of e-commerce technology and their business significance**
- **Describe the major types of e-commerce**
- **Understand the visions and forces behind the 1st E-Commerce era**

Learning Objectives

- **Understand the successes and failures of the 1st E-Commerce**
- **Identify several factors that will define the 2nd E-commerce era**
- **Describe the major themes underlying the study of e-commerce**
- **Identify the major academic disciplines contributing to e-commerce research**

Learning Objectives

- **Identify the key components of e-commerce business models.**
- **Describe the major B2C business models.**
- **Describe the major B2B business models.**
- **Recognize business models in other emerging areas of e-commerce.**
- **Understand key business concepts and strategies applicable to e-commerce.**

Amazon.com: Before and After

- Most well-known e-commerce company
- Conceived by Jeff Bezos in 1994
  - Opened in July 1995
- Four compelling reasons to shop
  - **Selection** (1.1 million titles at its opening time)
  - **Convenience** (anytime, anywhere)
  - **Price** (high discounts on bestsellers)
  - **Service** (one-click shopping, automated order confirmation, tracking, and shipping information)

Amazon.com: Before and After

Revenues and Earnings		
	Revenues	**Earnings**
1996	$15.6 Million	($6.24 Million)
1997	$148 Million	($31 Million)
1998	$610 Million	($125 Million)
1999	$1.6 Billion	($720 Million)
2000	$2.7 Billion	($1.4 Billion)
2008	**$19.16 Billion**	**$645 Million**

Losses

No profit until 2001: $5M

Amazon.com: Before and After

•Amazon annual/quarterly revenue history and growth rate from 2010 to 2023. Revenue can be defined as the amount of money a company receives from its customers in exchange for the sales of goods or services. Revenue is the top line item on an income statement from which all costs and expenses are subtracted to arrive at net income.

- Amazon revenue for the quarter ending September 30, 2023 was $143.083B, a 12.57% increase year-over-year.
- Amazon revenue for the twelve months ending September 30, 2023 was $554.028B, a 10.32% increase year-over-year.
- Amazon annual revenue for 2022 was $513.983B, a 9.4% increase from 2021.
- Amazon annual revenue for 2021 was $469.822B, a 21.7% increase from 2020.
- Amazon annual revenue for 2020 was $386.064B, a 37.62% increase from 2019.

E-commerce vs. E-business

E-commerce involves

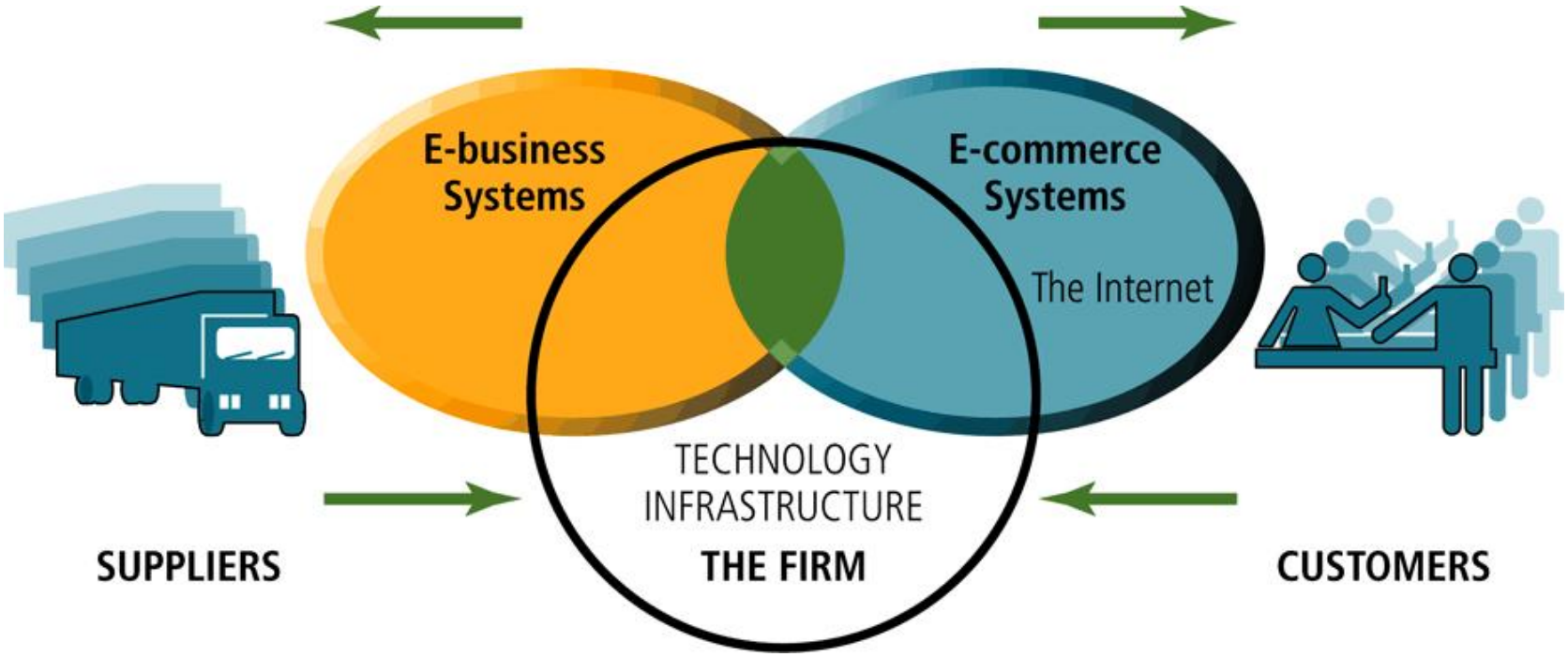
- **Digitally enabled commercial transactions between organizations and individuals.**
  - **Digitally enabled transactions include all transactions mediated by digital technology**
- **Commercial transactions involve the exchange of value across organizational or individual boundaries in return for products or services**

E-commerce vs. E-business

E-business involves

- **Digital enablement of transactions and processes *within* a firm, involving information systems under the control of the firm**
- **E-business does not involve commercial transactions across organizational boundaries where value is exchanged**

The Difference Between E-commerce and E-Business



Why Study E-commerce?

- E-commerce technology is different, more powerful than previous technologies (TV, Radio, Railway, Road etc.
- E-commerce brings fundamental changes to commerce
- Traditional commerce:
  - –Consumer as passive targets
  - –Mass-marketing driven
  - –Sales-force driven
  - –Fixed prices
  - –Information asymmetry:-
    - –*also known as "**information** failure," occurs when one party to an economic transaction possesses greater material knowledge than the other party. ... Almost all economic transactions involve **information asymmetries**.*

How are e-commerce transactions viewed

- They can be viewed from two perspectives:

a. Sell-side e-commerce-

   - in this view it refers to transactions involved in selling products to organizations customers

b. Buy-side e-commerce

   - refers to transactions involved in the procurement of resources needed by organizations from its suppliers.

Seven Unique Features of E-commerce Technology and Their Business Significance

TABLE 1.1 SEVEN UNIQUE FEATURES OF E-COMMERCE TECHNOLOGY

E-COMMERCE TECHNOLOGY DIMENSION	BUSINESS SIGNIFICANCE
Ubiquity — Internet/Web technology is available everywhere: at work, at home, and elsewhere via mobile devices, anytime.	The marketplace is extended beyond traditional boundaries and is removed from a temporal and geographic location. "Marketspace" is created; shopping can take place anywhere. Customer convenience is enhanced, and shopping costs are reduced.
Global Reach — The technology reaches across national boundaries, around the earth.	Commerce is enabled across cultural and national boundaries seamlessly and without modification. "Marketspace" includes potentially billions of consumers and millions of businesses worldwide.
Universal Standards — There is one set of technology standards, namely Internet standards.	There is one set of technical media standards across the globe.
Richness — Video, audio, and text messages are possible.	Video, audio, and text marketing messages are integrated into a single marketing message and consuming experience.
Interactivity — The technology works through interaction with the user.	Consumers are engaged in a dialog that dynamically adjusts the experience to the individual, and makes the consumer a co-participant in the process of delivering goods to the market.
Information Density — The technology reduces information costs and raises quality.	Information processing, storage, and communication costs drop dramatically, while currency, accuracy, and timeliness improve greatly. Information becomes plentiful, cheap, and accurate.
Personalization/Customization — The technology allows personalized messages to be delivered to individuals as well as groups.	Personalization of marketing messages and customization of products and services are based on individual characteristics.

Features Cont..

Social technology:

E-commerce and internet technologies allows users to create and share content in different forms (text, video and audio) in a world wide community. This brings out a social face of these technologies.

- **New internet and e-commerce technologies give the users power to create and distribute content on a large scale and allow users to program their own content consumption (who should consume)**
- **Internet creates a many-to-many model of mass communication**

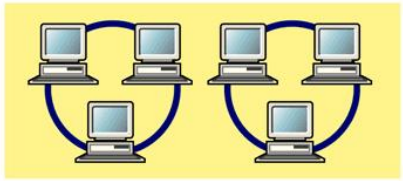
Technology upon which e-commerce Is built

- E-commerce is a technology driven phenomenon that relies on a host of information technologies-core technologies include internet and web
- Complementary technologies are computers , hand-held devices, LANs, Relational databases, client/server computing, fiber-optic switches etc.
- E-commerce relies on these technologies, not just internet
- Business applications on the other hand –i.e. the potential extraordinary returns on investment create interest and excitement in e-commerce.

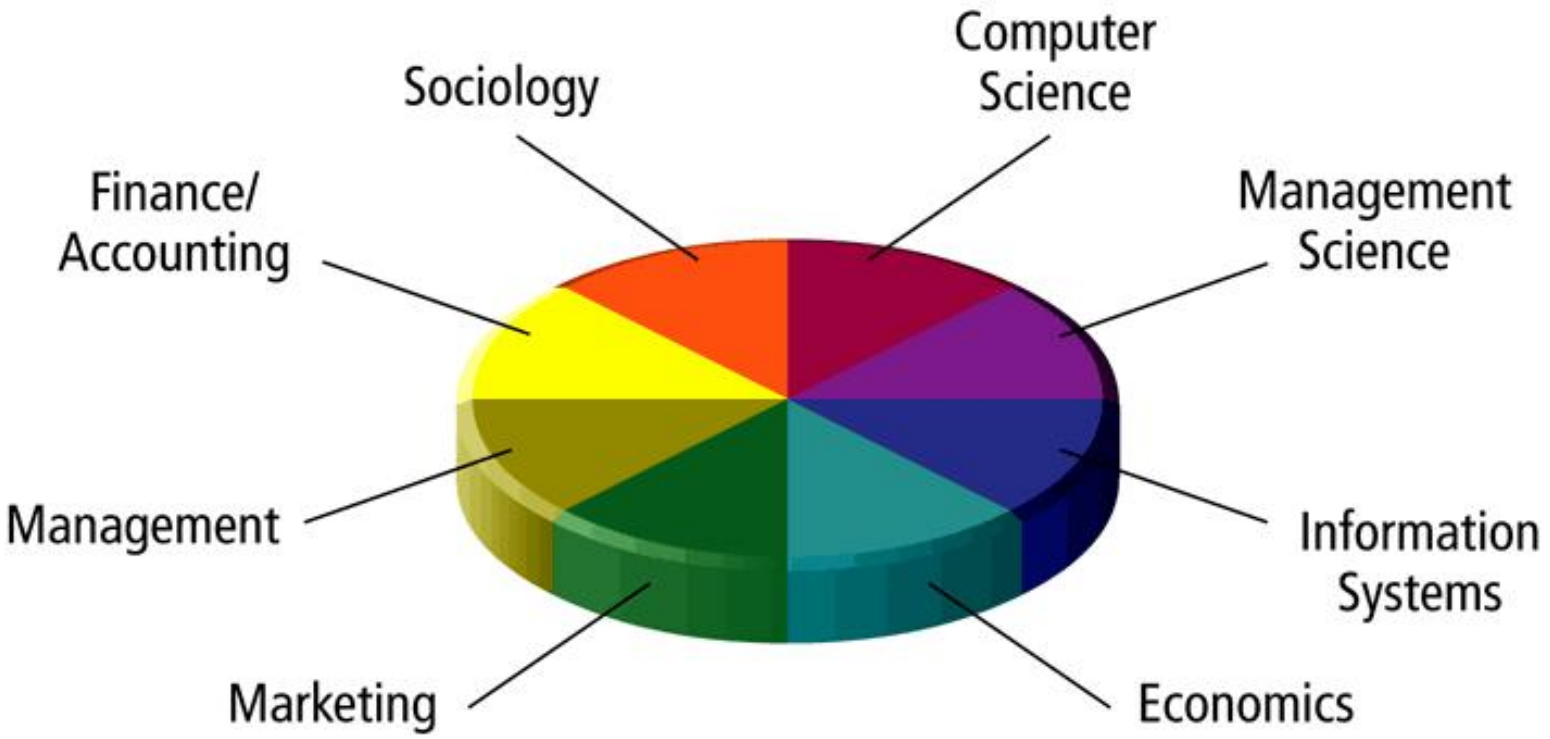
Web 2.0

- User-centered applications and social media technologies
  - –User-generated content and communication
  - –Highly interactive, social communities
  - –Large audiences; yet mostly unproven business models
  - –Examples:
    - – Twitter,
    - – YouTube,
    - – Instagram,
    - – TikTok
    - – Wikipedia,
    - – Tumblr

The Internet and the Evolution of Corporate Computing

COMPUTING TECHNOLOGY		BUSINESS APPLICATION
Mainframe Computers 1950 – 1975		Transaction automation Payroll Accounts receivable
Minicomputers 1970 – 1980		Business function automation Marketing Human Resources Design
Personal Computers 1980 – Present		Desktop automation Word processing Spreadsheets Databases
Local Area Networks Client/Server Computing 1980 – Present		Workgroup automation Document sharing Project management Messaging, e-mail
Enterprise-Wide Computing 1990 – Present		Enterprise-wide automation Resource planning systems Integrated finance-manufacturing systems Human resource planning
Internet and World Wide Web 1995 – Present		Industrial system automation Supply chain management Customer relationship management Channel management systems

Disciplines Concerned with E-Commerce



Academic Disciplines Concerned with E-commerce

- **Technical approach**
  - –Computer science
  - –Management science
  - –Information systems
- **Behavioral approach**
  - –Information systems
  - –Economics
  - –Marketing
  - –Management
  - –Finance/accounting
  - –Sociology

Major Types of E-Commerce

TABLE 1.2 MAJOR TYPES OF E-COMMERCE

TYPE OF E-COMMERCE	EXAMPLE
B2C — Business to Consumer	Amazon.com is a general merchandiser that sells consumer products to retail consumers.
B2B — Business to Business	eSteel.com is a steel industry exchange that creates an electronic market for steel producers and users.
C2C — Consumer to Consumer	eBay.com creates a marketspace where consumers can auction or sell goods directly to other consumers.
P2P — Peer to Peer	Gnutella is a software application that permits consumers to share music with one another directly, without the intervention of a market maker as in C2C e-commerce.
M-commerce — Mobile commerce	Wireless mobile devices such as PDAs (personal digital assistants) or cell phones can be used to conduct commercial transactions.

Major Types of E-Commerce

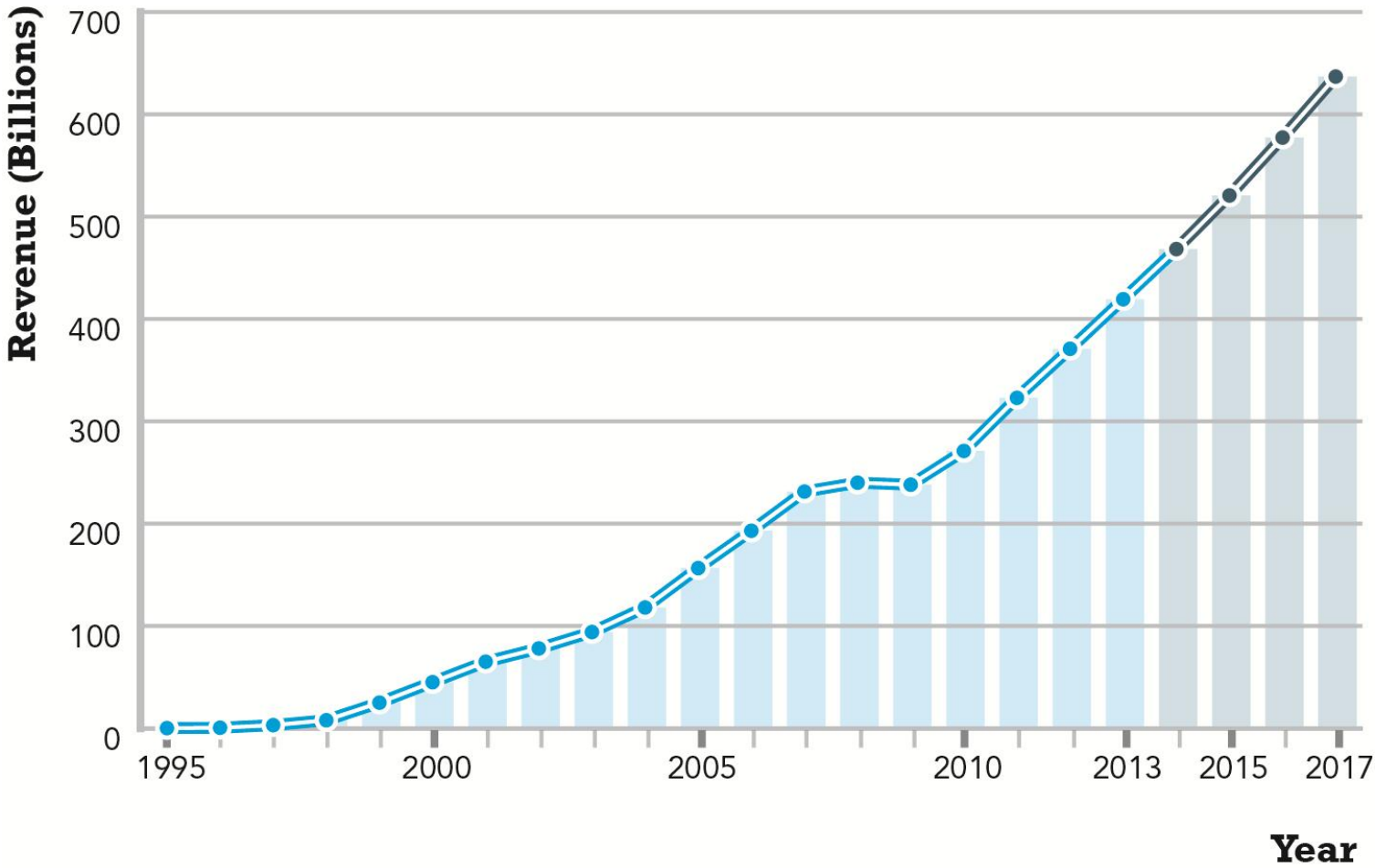
- **Market relationships**
  - **Business-to-Consumers (B2C)**
  - **Business-to-Business (B2B)**
  - **Consumer-to-Consumer (C2C)**
- **Technology-based**
  - **Peer-to-Peer (P2P)**
  - **Mobile Commerce (M-commerce)**

Business-to-Consumer E-commerce

- **Most commonly discussed type**
- **Online businesses attempt to reach individual consumers**

The Growth of B2C E-commerce

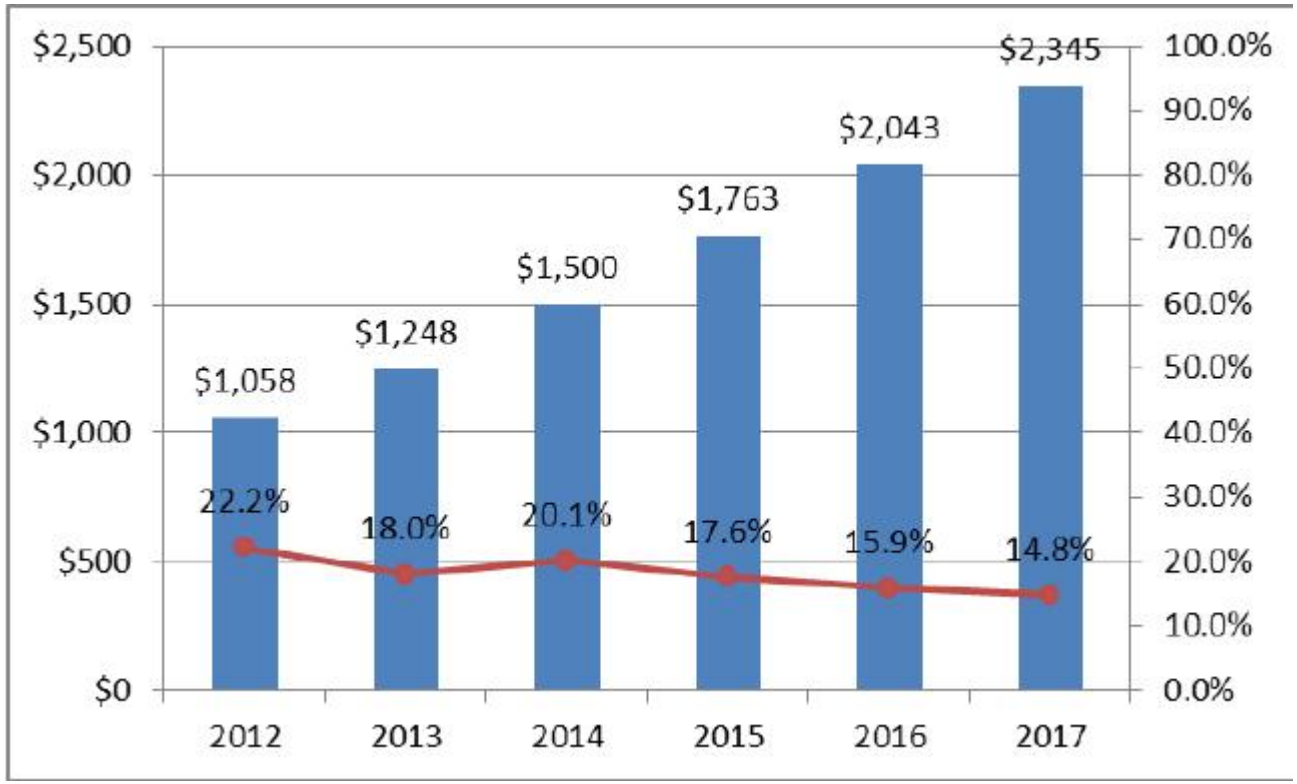
Figure 1.3, Page 20



SOURCE: Based on data from eMarketer, Inc., 2013a; authors' estimates.

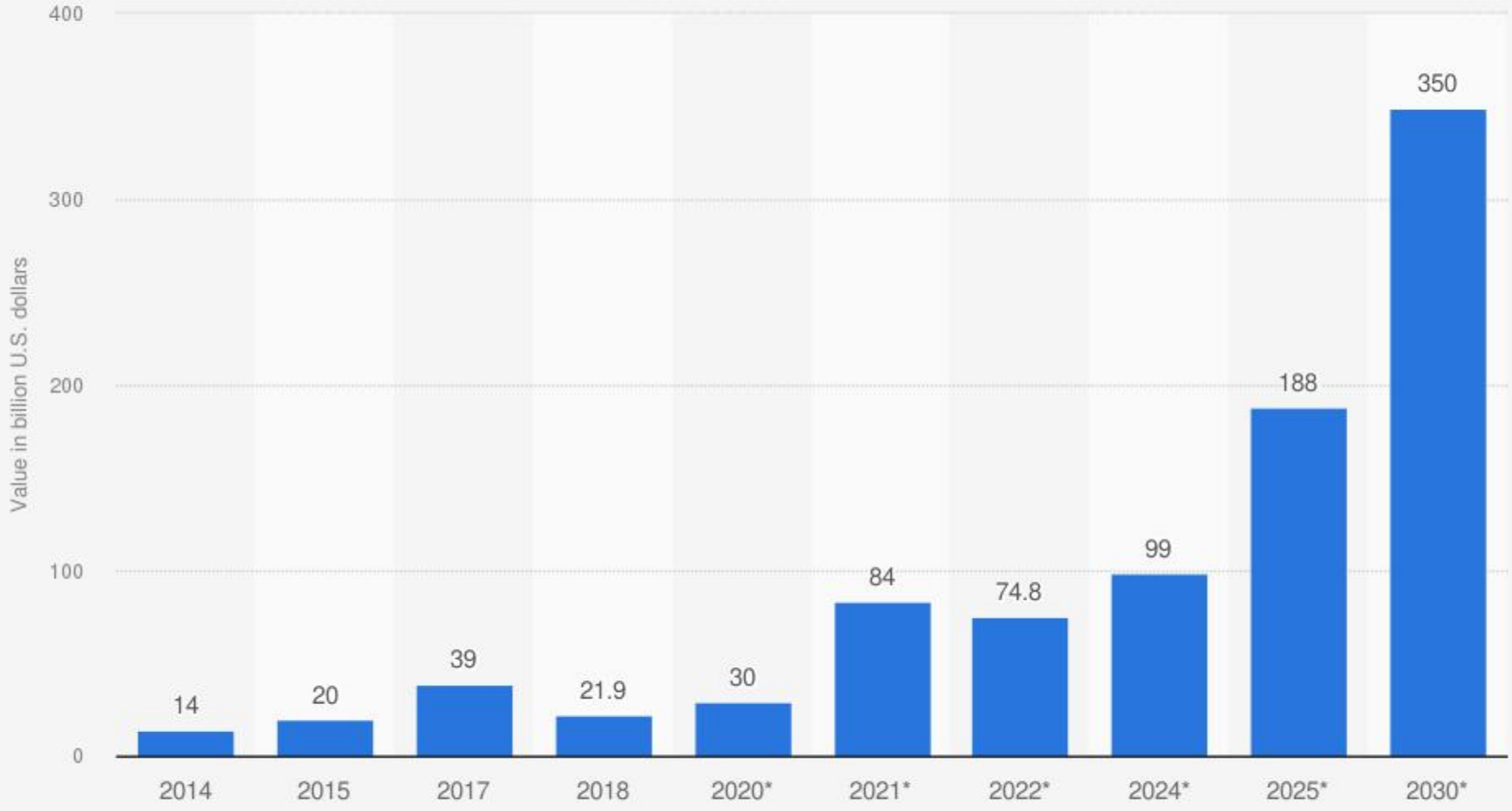
The Growth of B2C E-commerce

Figure 1.3, Page 20



SOURCE: Based on data from eMarketer, Inc., 2013a; authors' estimates.

Market size of e-commerce industry across India from 2014 to 2018, with forecasts until 2030 (in billion U.S. dollars)



Sources
India Brand Equity Foundation; Various sources (Media sources); BCG; Bain & Company; Morgan Stanley

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Additional Information:
Various sources (Media sources); BCG; Bain & Company; Morgan Stanley; 2014 to 2018

The Growth of B2C E-commerce

Figure 1.3, Page 20



SOURCE: Based on data from eMarketer, Inc., 2013a; authors' estimates.

Business-to-Business E-commerce

- Businesses focus on sell to other businesses
- Largest form of e-commerce
- Primarily involved inter-business exchanges at first
- Other models have developed
  - e-distributors
  - infomediaries
  - B2B service providers

Consumer-to-Consumer E-commerce

- **Provide a way for consumers to sell to each other**
- **Estimated $5 billion market**
- **Consumer:**
  - **prepares the product for market**
  - **places the product for auction or sale**
  - **relies on market maker to provide catalog, search engine, and transaction clearing capabilities**

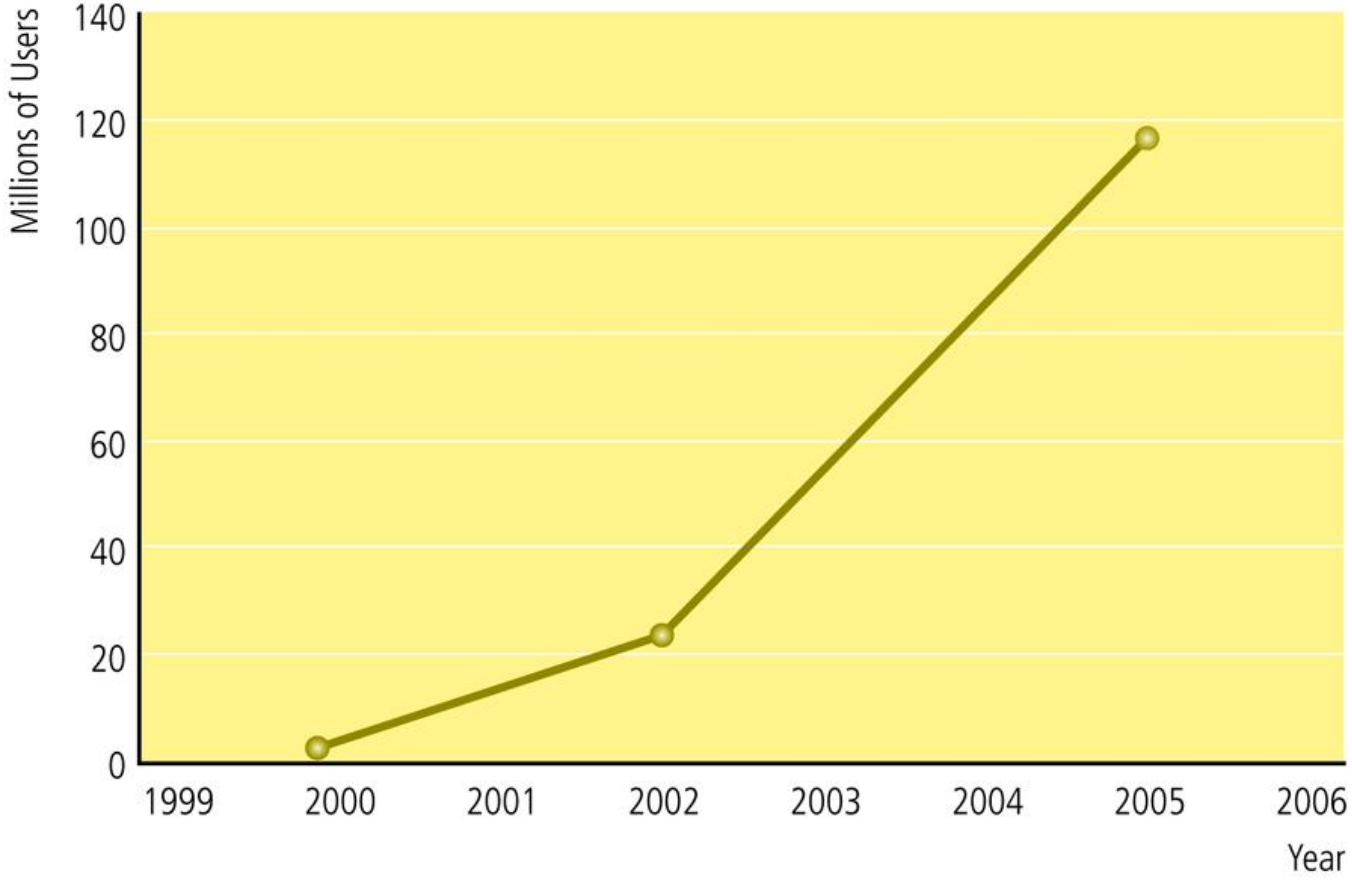
Peer-to-Peer E-commerce

- **Enables Internet users to share files and computer resources**
- **Napster (early example)**
- **Skype (more modern and successful example)**

Mobile E-commerce

- **Wireless digital devices enable transactions on the Web**
- **Uses personal digital assistants (PDAs) to connect**
- **Used most widely in Japan and Europe**
- **Growth in Kenya**

Web Access Via Wireless Devices in the United States



Kenyan statistics

- A whopping 99% of the internet traffic in Kenya is done via mobile operators, meaning 3G, Edge or GPRS.-2010

Table 7: Internet subscriptions by operator

S/No.	Name of Operator	Subscriptions	Market share (%)
1.	Safaricom Ltd	2,977,584	92.18
2.	Celtel Kenya Ltd	149,053	4.61
3.	Telkom Orange	66,030	2.40
	Telkom Fixed	11,638	
4.	Communication Solutions(Access Kenya) Ltd	7,512	0.23
5.	Wananchi Telkom Ltd	7,500	0.23
6.	Kenya Data Networks	5,451	0.17
7.	Africa Online	1,608	0.05
8.	Flexible Bandwidth	1,198	0.04
9.	Swift Global	1,133	0.04
10.	Callkey Networks Ltd	800	0.02
11.	Others	516	0.02

Source: CCK, Operators' returns

Technology and E-Commerce in Perspective

Although e-commerce has grown explosively, there is no guarantee it will continue to grow

E-Commerce I and II

- **E-Commerce I (1995-2000)**
  - **Explosive growth starting in 1995**
  - **Widespread of Web to advertise products**
  - **Ended in 2000 when dot.com began to collapse**
- **E-Commerce II (2001-2006)**
  - **Began in January 2001**
  - **Reassessment of e-commerce companies**

E-Commerce II 2001-2006

- **Crash in stock market values of E-commerce I companies throughout 2000 is an end to E-commerce I**
- **Led to a sobering reassessment of the prospects of e-commerce and the methods of achieving business success.**
- **E-commerce II begins in 2001 and ends five year later -- the limit for making technology and business projections**

E-Commerce II 2001-2006

- **Reasons for the end of E-Commerce I**
  - run-up in technology stocks due to enormous information technology capital expenditure of firms rebuilding their internal business systems to withstand Y2K
  - telecommunications industry had built excess capacity in high-speed fiber optic networks
  - 1999 e-commerce Christmas season provided less sales growth that anticipated and demonstrated e-commerce was not easy (eToys.com)
  - valuations of technology companies had risen so high supporters were questioning whether earnings could justify the prices of the shares.

E-Commerce I and E-Commerce II Compared Innovation wise

TABLE 1.5 E-COMMERCE I AND E-COMMERCE II COMPARED

E-COMMERCE I	E-COMMERCE II
Technology-driven	Business-driven
Revenue growth emphasis	Earnings and profits emphasis
Venture capital financing	Traditional financing
Ungoverned	Stronger regulation and governance
Entrepreneurial	Large traditional firms
Disintermediation	Strengthening intermediaries
Perfect markets	Imperfect markets, brands, and network effects
Pure online strategies	Mixed "clicks and bricks" strategies
First mover advantages	Strategic follower strength

E-Commerce Business Models

- **Business model**
  - **a set of planned activities designed to result in a profit in a marketplace**
- **E-commerce business model**
  - **a business model that aims to use and leverage the unique qualities of the Internet and the World Wide Web.**

Eight Key Ingredients of a Business Model

Table 2.1

TABLE 2.1 KEY INGREDIENTS OF A BUSINESS MODEL

BUSINESS MODEL COMPONENTS	KEY QUESTIONS
Value proposition	Why should the customer buy from you?
Revenue model	How will you earn money?
Market opportunity	What marketspace do you intend to serve, and what is its size?
Competitive environment	Who else occupies your intended marketspace?
Competitive advantage	What special advantages does your firm bring to the marketspace?
Market strategy	How do you plan to promote your products or services to attract your target audience?
Organizational development	What types of organizational structures within the firm are necessary to carry out the business plan?
Management team	What kinds of experiences and background are important for the company's leaders to have?

Eight Key Ingredients of a Business Model: 1.Value Proposition

- **Defines how a company's product or service fulfills the needs of customers.**
- **Questions**
  - **Why will customers choose to do business with your firm instead of another company?**
  - **What will your firm provide that other firms do not and cannot?**

Eight Key Ingredients of a Business Model: 2.Revenue Model

- **Describes how the firm will earn revenue, produce profits, and produce a superior return on invested capital.**
- **E-commerce revenue models include:**
  - **advertising model**
  - **subscription model**
  - **transaction fee model**
  - **sales model**
  - **affiliate model**

Eight Key Ingredients of a Business Model: Revenue Model

- **Advertising revenue model**
  - **a company provides a forum for advertisements and receives fees from advertisers (Yahoo)**
- **Subscription revenue model**
  - **a company offers it users content or services and charges a subscription fee for access to some or all of it offerings (Consumer Reports or Wall Street Journal)**

Eight Key Ingredients of a Business Model: Revenue Model

- **Transaction fee revenue model**
  - a company receives a fee for enabling or executing a transaction (eBay or E-Trade)
- **Sales revenue model**
  - a company derives revenue by selling goods, information, or services (Amazon or DoubleClick)
- **Affiliate revenue model**
  - a company steers business to an affiliate and receives a referral fee or percentage of the revenue from any resulting sales (MyPoints)

Five Primary Revenue Models

Table 2.2

TABLE 2.2 FIVE PRIMARY REVENUE MODELS

REVENUE MODEL	EXAMPLE	REVENUE SOURCE
Advertising	Yahoo.com	Fees from advertisers in exchange for advertisements
Subscription	WSJ.com, Consumerreports.org, Sportsline.com	Fees from subscribers in exchange for access to content or services
Transaction Fee	eBay.com, E-Trade.com	Fees (commissions) for enabling or executing a transaction
Sales	Amazon.com, DoubleClick.net, Salesforce.com	Sales of goods, information, or services
Affiliate	MyPoints.com	Fees for business referrals

Eight Key Ingredients of a Business Model: 3.Market Opportunity

- **Market opportunity**
  - **refers to the company's intended marketspace and the overall potential financial opportunities available to the firm in that market space**
  - **defined by the revenue potential in each of the market niches where you hope to compete**
- **Marketspace**
  - **the area of actual or potential commercial value in which a company intends to operate**

Eight Key Ingredients of a Business Model: 4.Competitive Environment

- **Refers to the other companies operating in the same marketplace selling similar products**
- **Influenced by:**
  - **how many competitors are active**
  - **how large are their operations**
  - **the market share of each competitor**
  - **how profitable these firms are**
  - **how they price their products**

Eight Key Ingredients of a Business Model: 5.Competitive Advantage

- **Achieved by a firm when it can produce a superior product and/or bring the product to market at a lower price than most, or all, of its competitors**
- **Achieved because a firm has been able to obtain differential access to the factors of production that are denied their competitors -- at least in the short term**

Eight Key Ingredients of a Business Model: Competitive Advantage Concepts

- **Asymmetry**
  - **exists whenever one participant in a market has more resources than other participants**
- **First mover advantage**
  - **a competitive market advantage for a firm that results from being the first into a marketplace with a serviceable product or service**

Eight Key Ingredients of a Business Model: Competitive Advantage

- **Unfair competitive advantage**
  - **occurs when one firm develops an advantage based on a factor that other firms cannot purchase**
- **Perfect Market**
  - **a market in which there are no competitive advantages or asymmetries because all firms have equal access to all the factors of production**
- **Leverage**
  - **when a company uses its competitive advantage to achieve more advantage in surrounding markets**

Eight Key Ingredients of a Business Model: 6.Market Strategy

- **The plan you put together that details exactly how you intend to enter a new market and attract new customers**
- **Best business concepts will fail if not properly marketed to potential customers**

Eight Key Ingredients of a Business Model: 7. Organizational Development

- **Describes how the company will organize the work that needs to be accomplished**
- **Work is typically divided into functional departments**
- **Move from generalists to specialists as the company grows**

Eight Key Ingredients of a Business Model: 8. Management Team

- **Employees of the company responsible for making the business model work**
- **Strong management team gives instant credibility to outside investors**
- **A strong management team may not be able to salvage a weak business model**
- **Should be able to change the model and redefine the business as it becomes necessary**

Major Business-to-Consumer (B2C) Business Models

Page 67, Table 2.3

TABLE 2.3 B2C BUSINESS MODELS

BUSINESS MODEL	VARIATIONS	EXAMPLES	DESCRIPTION	REVENUE MODEL
Portal	Horizontal/General	Yahoo.com, AOL.com, MSN.com, Excite@home.com	Offers an integrated package of services and content such as search, news, e-mail, chat, music downloads, video streaming, and calendars. Seeks to be a user's home base.	Advertising, subscription fees, transaction fees
	Vertical/Specialized (Vortal)	iBoats.com	Offers services and products to specialized marketspace.	Advertising, subscription fees, transaction fees
E-tailer	Virtual Merchant	Amazon.com	Online version of retail store, where customers can shop at any hour of the day or night without leaving home or office.	Sales of goods
	Clicks and Mortar	Walmart.com	Online distribution channel for company that also has physical stores.	Sales of goods
	Catalog Merchant	LandsEnd.com	Online version of direct mail catalog.	Sales of goods
	Online Mall	Fashionmall.com	Online version of mall.	Sales of goods, transaction fees
	Manufacturer-direct	Dell.com	Online sales made directly by manufacturer.	Sales of goods
Content Provider		WSJ.com, Sportsline.com, CNN.com	Information and entertainment providers such as newspapers, sports sites, and other online sources that offer customers up-to-date news and special interest, how-to guidance, and tips and/or information sales.	Advertising, subscription fees, affiliate referral fees

(continued)

Major Business-to-Consumer (B2C) Business Models

Page 68, Table 2.3 *continued*

TABLE 2.3 B2C BUSINESS MODELS (*CONTINUED*)

BUSINESS MODEL	VARIATIONS	EXAMPLES	DESCRIPTION	REVENUE MODEL
Transaction Broker		E-Trade.com, Expedia.com, Monster.com	Processors of online sales transactions, such as stock brokers and travel agents, that increase customers' productivity by helping them get things done faster and more cheaply.	Transaction fees
Market Creator	Auctions and other forms of dynamic pricing	eBay.com, Priceline.com	Web-based businesses that use Internet technology to create markets that bring buyers and sellers together.	Transaction fees
Service Provider		xDrive.com, whatsitworthtoyou.com, myCFO.com	Companies that make money by selling users a service, rather than a product.	Sales of services
Community Provider		About.com, iVillage.com, BlackPlanet.com	Sites where individuals with particular interests, hobbies, and common experiences can come together and compare notes.	Advertising, subscription, affiliate referral fees

Major Business-to-Consumer (B2C) Business Models

- **Portal**
  - **offers powerful search tools plus an integrated package of content and services**
  - **typically utilizes a combines subscription/advertising revenues/transaction fee model**
  - **may be general or specialize (vortal)**

Major Business-to-Consumer (B2C) Business Models

- **E-tailer**
  - **online version of traditional retailer**
  - **includes**
    - **virtual merchants (online retail store only)**
    - **clicks and mortar e-tailers (online distribution channel for a company that also has physical stores)**
    - **catalog merchants (online version of direct mail catalog)**
    - **online malls (online version of mall)**
    - **Manufacturers selling directly over the Web**

Major Business-to-Consumer (B2C) Business Models

- **Content Provider**
  - **information and entertainment companies that provide digital content over the Web**
  - **typically utilizes an advertising, subscription, or affiliate referral fee revenue model**
- **Transaction Broker**
  - **processes online sales transactions**
  - **typically utilizes a transactions fee revenue model**

Major Business-to-Consumer (B2C) Business Models

- **Market Creator**
  - uses Internet technology to create markets that bring buyers and sellers together
  - typically utilizes a transaction fee revenue model
- **Service Provider**
  - offers services online
- **Community Provider**
  - provides an online community of like-minded individuals for networking and information sharing
  - revenue is generated by referral fee, advertising, and subscription

Insight on Technology: Goggle.com -- Searching for Profits

- **Web's hottest search engine**
- **Started in 1998 by two enterprising Stanford grad students**
- **Uses outside criteria to validate that a search result is likely to be relevant**
  - **the more outside links there are to a particular page, the higher it jumps in Google's ranking structure**

Major Business-to-Business (B2B) Business Models

Table 2.4

TABLE 2.4 B2B BUSINESS MODELS

BUSINESS MODEL	VARIATIONS	EXAMPLES	DESCRIPTION	REVENUE MODEL
Marketplace/ Exchange (B2B Hub)	Vertical	DirectAg.com, e-Steel.com	Helps bring buyers and sellers together to reduce procurement costs for a specific industry.	Transaction fees
	Horizontal	TradeOut.com	Same as vertical except focused on specific types of products and services.	Transaction fees
E-Distributor		Grainger.com	Connecting businesses directly with other businesses, reducing sales cycles and mark-up.	Sales of goods
B2B Service Provider	Traditional	Employeematters.com	Supports companies through online business services.	Sales of services
	Application Service Provider (ASP)	Salesforce.com, Corio.com	Rents Internet-based software applications to businesses.	Rental fees
Matchmaker		iShip.com	Helps businesses find what they want and need on the Web.	Transaction fees
Infomediary	Audience Broker	DoubleClick.net	Gathers information about consumers and uses it to help advertisers find the most appropriate audience.	Sales of information
	Lead Generator	AutoByTel.com	Gathers customer data, and uses it to direct vendors to customers.	Referral fee

Major Business-to-Business (B2B) Business Models

- **B2B Hub**
  - **also known as *marketplace/exchange***
  - **electronic marketplace where suppliers and commercial purchasers can conduct transactions**
  - **may be a general (horizontal marketplace) or specialized (vertical marketplace)**
- **E-distributor**
  - **supplies products directly to individual businesses**

Major Business-to-Business (B2B) Business Models

- **B2B Service Provider**
  - sells business services to other firms
- **Matchmaker**
  - links businesses together
  - charges transaction or usage fees
- **Infomediary**
  - gather information and sells it to businesses

Insight on Business: E-Steel.com Breaks the Mold

- **B2B marketplace**
- **3,500 member companies trading globally**
- **Uses private negotiation model rather than auction model**

Business Models in Other Emerging Areas of E-Commerce

Table 2.5

TABLE 2.5 BUSINESS MODELS IN EMERGING E-COMMERCE AREAS

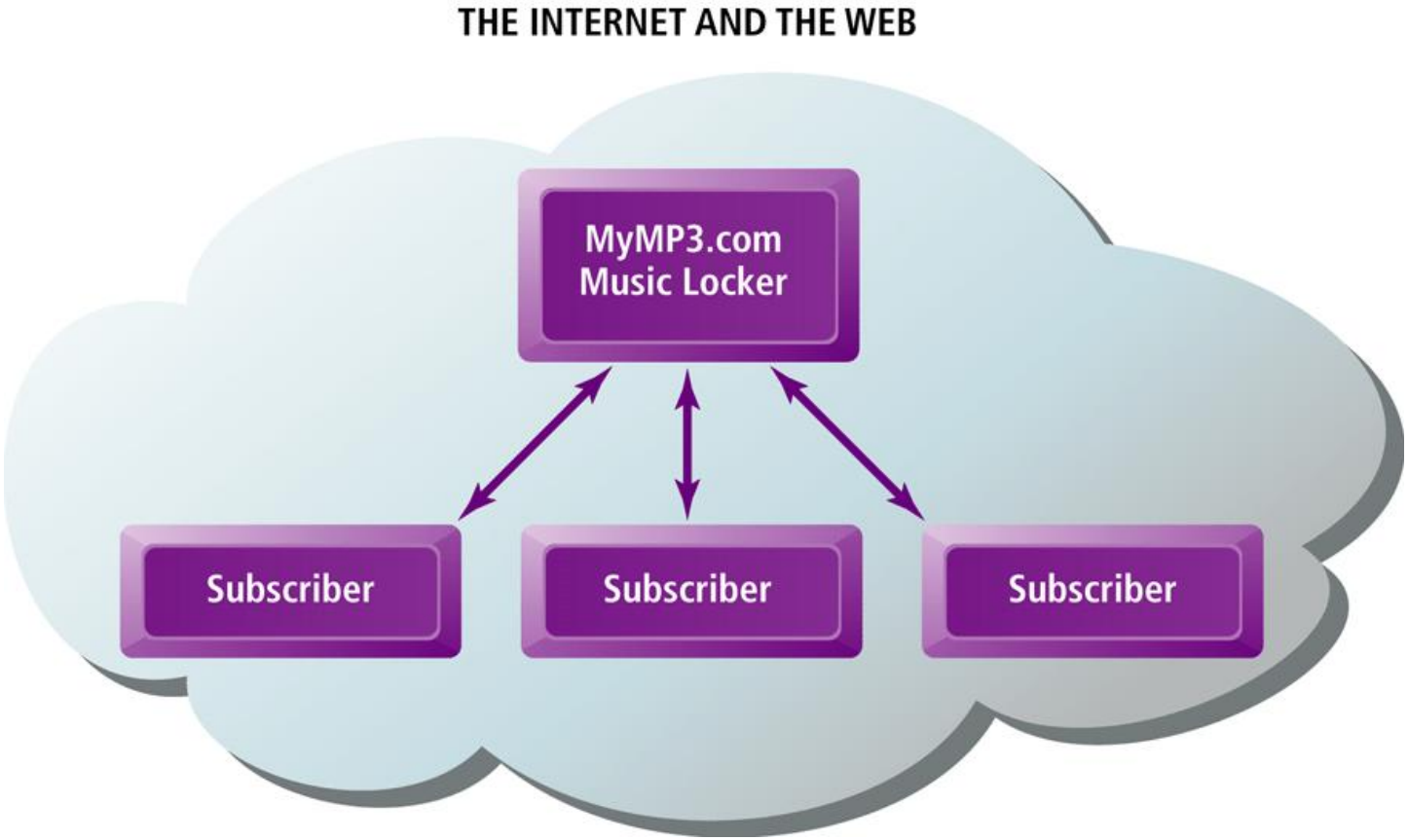
TYPE	MODEL	EXAMPLE	DESCRIPTION	REVENUE MODEL
Consumer-to-consumer	Market Creator	eBay.com, Half.com	Helps consumers connect with other consumers who have items to sell.	Transaction fees
Peer-to-peer	Content Provider	Napster.com, My.MP3.com	Technology enabling consumers to share files and services via the Web.	Subscription fees, advertising, transaction fees
M-commerce	Various	Amazon.com	Extending business applications using wireless technology.	Sales of goods

Business Models in Other Emerging Areas of E-Commerce

- **C2C Business Models**
  - **connect consumers with other consumers**
  - **most successful has been the market creator business model**
- **P2P Business Models**
  - **enable consumers to share file and services via the Web without common servers**
  - **a challenge to find a revenue model that work**
    - **Skype !!**

Business Models in Other Emerging Areas of E-Commerce

Figure 2.2



Business Models in Other Emerging Areas of E-Commerce

- **M-commerce Business Models**
  - **traditional e-commerce business models leveraged for emerging wireless technologies to permit mobile access to the Web**
- **E-commerce Enablers' Business Models**
  - **focus on providing infrastructure necessary for e-commerce companies to exist, grow, and prosper**

E-commerce Enablers

Table 2.6

TABLE 2.6 E-COMMERCE ENABLERS

INFRASTRUCTURE	PLAYERS
Hardware: Web Servers	IBM, Sun, Compaq, Dell
Software: Operating Systems and Server Software	Microsoft, Sun, Apache Software Foundation
Networking: Routers	Cisco
Security: Encryption Software	CheckPoint, VeriSign
E-commerce Software Systems (B2C, B2B)	IBM, Microsoft, iPlanet, CommerceNet, Ariba
Streaming Media Solutions	Real Networks, Microsoft
Customer Relationship Management Software	PeopleSoft
Payment Systems	PayPal, CyberCash
Performance Enhancement	Akamai, Cache Flow, Inktomi, Cidera, Digital Island
Databases	Oracle, Sybase
Hosting Services	Exodus, Equinex, Global Crossing

THE END

QUE & ANS

COMMENTS